LUMI

AI-Powered Elderly Companion

Empowering elderly care through emotion-aware AI assistance — detecting distress, monitoring wellbeing, and keeping families connected in real time.

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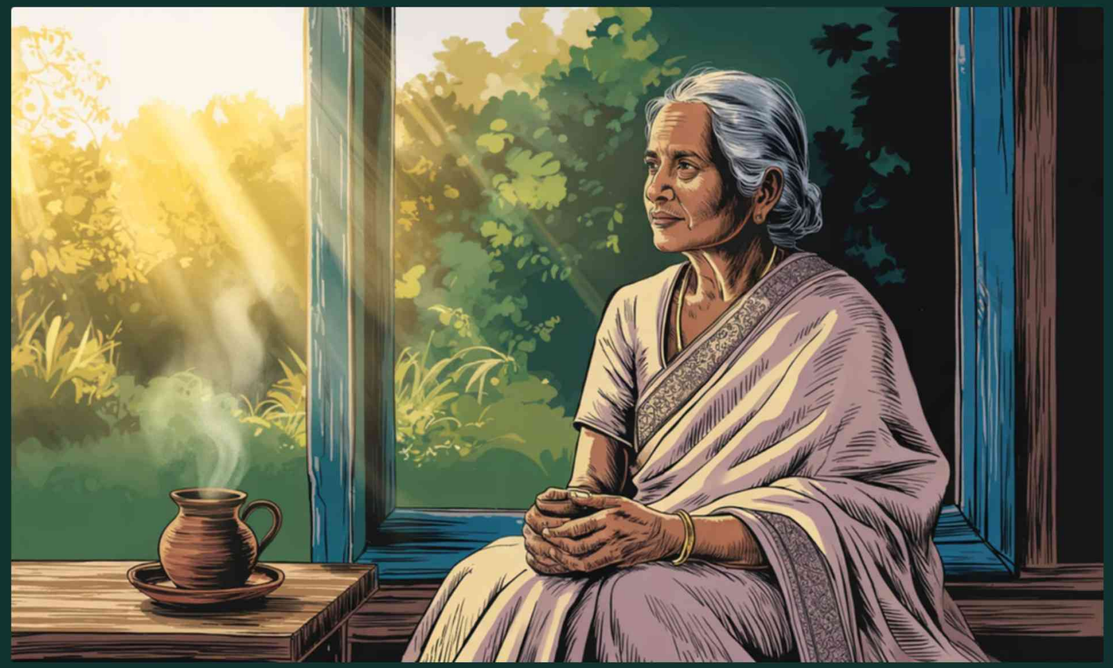
The Problem We're Solving



Isolation Is Invisible

Elderly people living alone face emotional isolation, delayed medical assistance, unnoticed falls, and a lack of continuous monitoring — often without anyone knowing until it's too late.

> LUMI bridges this gap with AI-powered emotional monitoring, emergency detection, and real-time family connectivity.

Meet Lakshmi Devi

72 years old · Bangalore · Retired Teacher · Lives Alone

Needs emotional support, medication reminders, and safety monitoring — but has no one checking in consistently.

Introducing LUMI

LUMI is an AI-powered elderly companion device that uses **Raspberry Pi, computer vision, voice AI, and sentiment analysis** to continuously monitor emotional wellbeing and deliver real-time alerts to families and caregivers.

🎙️ Voice AI

Converses naturally in Kannada, Hindi, and English

👁️ Computer Vision

Detects falls and physical distress in real time

💡 Sentiment Analysis

Understands emotional state through voice and expression

Why LUMI Stands Out

Five core capabilities that work together to protect and support elderly users around the clock.

AI Sentiment Analysis

Detects emotional state through voice tone and facial expression

Fall & Distress Detection

Computer vision identifies falls and physical emergencies instantly

Multilingual Voice

Supports Kannada, Hindi, and English for natural conversation

Firebase Real-Time Alerts

Instant notifications sent to family and caregivers when needed

Family Dashboard

Live wellbeing monitoring accessible to loved ones anywhere

Thinking Big: The Road Ahead

Where We Are Today

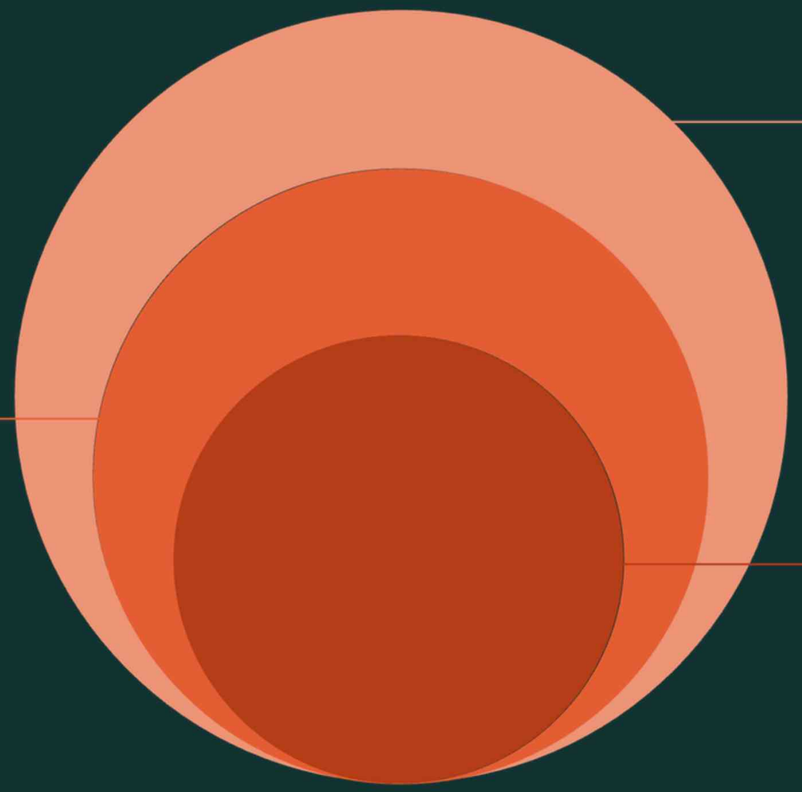
LUMI is a **functional prototype** — a Raspberry Pi-based monitoring system with working voice AI, sentiment analysis, fall detection, and Firebase alerts already demonstrated.

Where We're Headed

A **full AI healthcare ecosystem** — predictive care, smart home automation, clinical integration, and a scalable platform that transforms how elderly care is delivered across India and beyond.

System Architecture

LUMI's pipeline flows from physical sensors through on-device AI to cloud-based alerts — all in under 500ms.



Each layer is designed for low-latency, privacy-first operation — sensitive processing happens on-device before anything reaches the cloud.

How Data Flows Through LUMI

1

Capture

Camera and microphone record voice and video continuously

2

Process

AI models analyze sentiment, expression, and movement

3

Assess

Distress and fall detection algorithms evaluate risk level

4

Alert

Firebase triggers real-time notifications to family dashboard

5

Monitor

Family and caregivers view live wellbeing status anytime

Rollout Roadmap

1

Phase 1 — Prototype

Validated hardware and AI pipeline with core detection features

2

Phase 2 — Healthcare Integration

Connect with clinics, pharmacies, and caregiver networks

3

Phase 3 — Predictive AI

Health forecasting and early-warning intelligence for proactive care

4

Phase 4 — Smart Ecosystem

Full smart-home integration for autonomous elderly living support

Built for Everyone. Designed with Care.

Inclusivity & Fairness

- **Multilingual:** Kannada, Hindi, and English support
- **Voice-first:** No touchscreen or smartphone required
- **Low-cost:** Affordable Raspberry Pi hardware
- **Privacy-first:** Edge AI keeps sensitive data on-device

Technical Specifications

Hardware

Raspberry Pi 4B, Camera v2, PIR Sensor

AI Models

DeepFace · Whisper · MediaPipe Pose · Llama 3.1

Backend

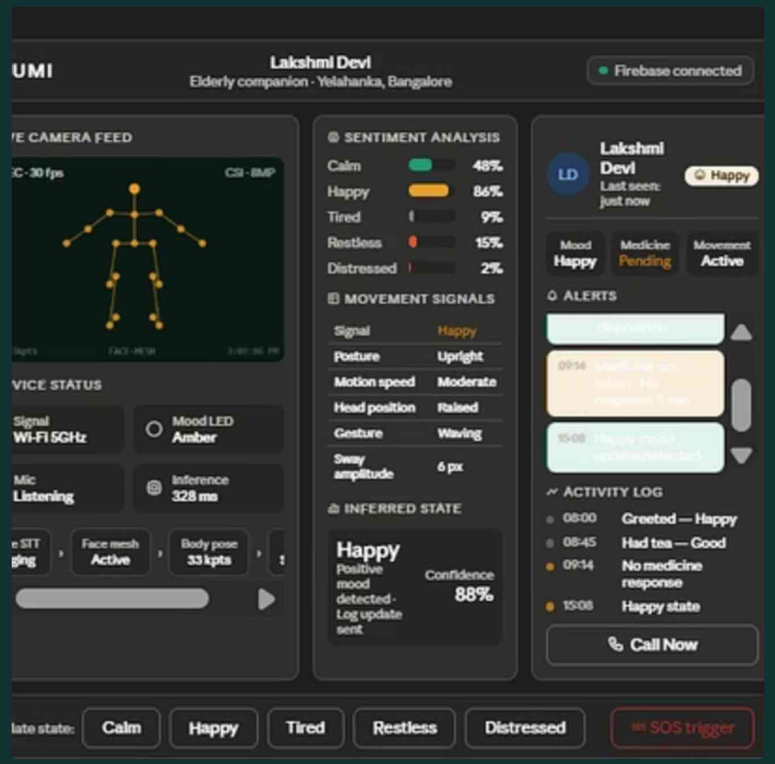
Firebase real-time database

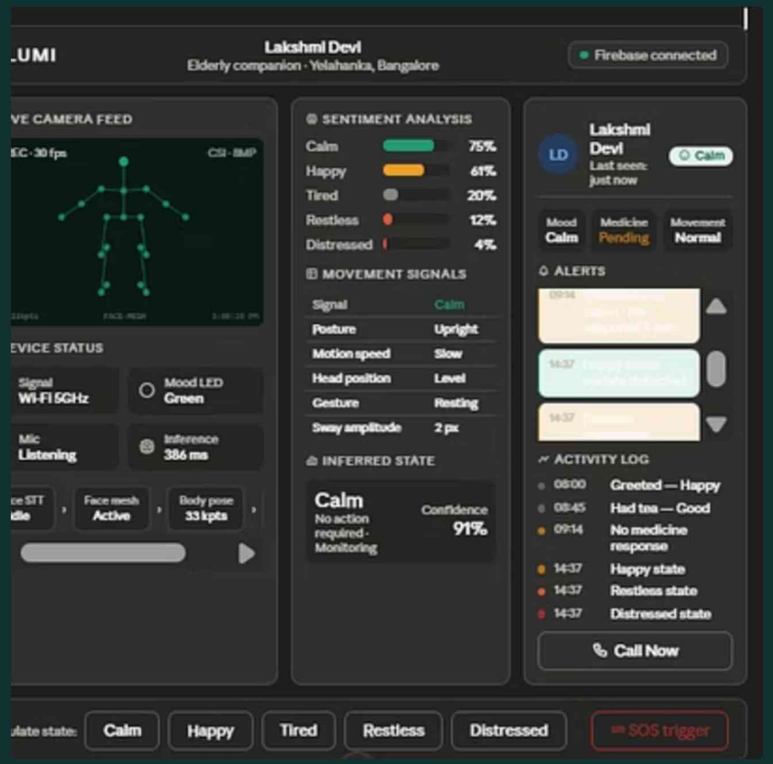
Performance

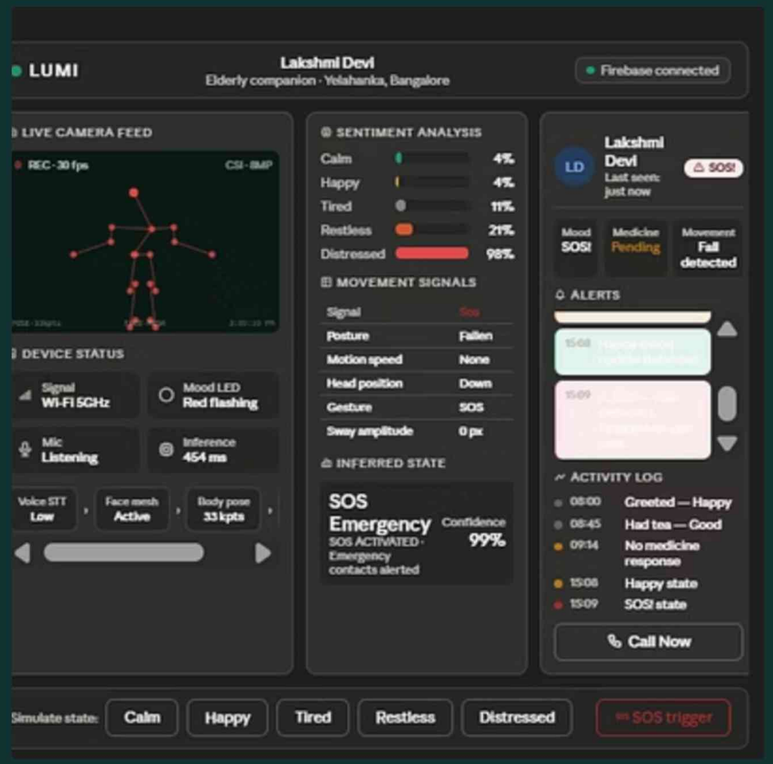
Under 500ms inference latency

LUMI in Action

Here's a look at our working prototype — the hardware, the interface, and the real-world setup bringing LUMI to life.







LUMI is ready for pilot partnerships. We're seeking healthcare providers, eldercare facilities, and family decision-makers to help us test and scale this solution in real-world settings.